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Ensuring Functional Equivalence in Retrofitted Anti-Skid Systems for European Public Transportation:A Hybrid Two-Layer Verification Methodology

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Ensuring Functional Equivalence in Retrofitted Anti-Skid Systems for European Public Transportation: A Hybrid Two-Layer Verification Methodology

Mert Torun

***Abstract*—Modernising ageing safety-critical electronics in public transportation is often more cost-effective and environmentally sustainable than complete system replacement—provided functional equivalence with the original design can be rigorously demonstrated under contemporary safety standards. This paper presents a case study of retrofitting a legacy anti-skid test-diagnosis module used in European urban rail. An obsolete Intel MCS-48 microcontroller was replaced with a flash-based Field-Programmable Gate Array (FPGA; Actel A3P1000) emulating the original processor and I/O logic, complemented by a galvanically isolated STM32 microcontroller subsystem for enhanced diagnostics. To verify the retrofit, a two-layer verification methodology was developed, derived from the V-model as defined in IEC 61508 and IEEE 1012: (1) deterministic component and interface testing via VHDL simulation and hardware timing measurements, and (2) scenario-based system-level testing using a dedicated hardware fixture with YAML-defined test cases. Through convergence of three independent evidence lines—simulation, oscilloscope measurement, and integrated scenario testing—functional equivalence of the safety-critical core was demonstrated for all documented fault codes, self-test sequences, and user interactions. Results were mapped to IEC 61508, EN 50129, EN 50716, and IEEE 1012. The methodology is assessed for transferability to other safety-critical retrofit contexts.**

***Index Terms*—Anti-Skid Protection, FPGA Emulation, Functional Equivalence, Functional Safety, IEC 61508, Retrofit Verification, Safety-Critical Systems, V-Model, Verification and Validation**

I. Introduction

MODERNISING ageing safety-critical systems in public transportation presents a dual challenge: operators must simultaneously extend the operational life of proven infrastructure and comply with increasingly stringent safety standards [1], [2]. As key components become obsolete, complete system replacement is often prohibitively expensive. Targeted retrofitting offers a compelling alternative—provided *functional equivalence* with the original system can be rigorously certified [3], [4].

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This paper addresses these challenges through a case study of a legacy anti-skid protection system [5], widely deployed across European urban rail networks. The core controller, based on the obsolete Intel MCS-48 microcontroller family, was replaced with a modern Field-Programmable Gate Array (FPGA)-based emulation augmented by an isolated diagnostics subsystem. While prior work has addressed formal verification of new safety-critical designs [6], [7] and Hardware-in-the-Loop (HIL) testing of complete braking systems [8], no established methodology exists for systematically verifying functional equivalence at the board level in legacy-to-FPGA retrofit scenarios. This paper addresses that gap.

Research question. How can functional equivalence in safety-critical hardware retrofits be rigorously demonstrated through structured, standards-aligned verification?

Contributions. The scientific and engineering contributions of this work are threefold:

(a) **Characterisation of a Verification Blind Spot:** We identify and formulate a fundamental diagnostic blind spot in board-level safety retrofits. We demonstrate that component-level simulation (white-box) lacks visibility into physical-layout defects and real-time execution overheads of decoupled modules, whereas system-level functional testing (black-box) is constrained by observational limits, failing to detect latent state anomalies (such as inactive memory interfaces). We prove that relying on either paradigm in isolation compromises safety-critical assurance.

(b) **A Layered Evidence-Convergence Methodology:** We propose a structured, standards-aligned (IEC 61508 / EN 50716) two-layer verification framework. By establishing mathematical and temporal correspondence between pre-synthesis simulation, JTAG boundary-scan, physical measurements, and automated, YAML-defined system-level scenario execution, the framework systematically closes the diagnostic blind spot.

(c) **Empirical Proof via Bug Classification:** We provide a detailed taxonomic classification of five distinct classes of hardware, firmware, and logical integration defects discovered during validation. This serves as an empirical validation of the diagnostic capabilities

of our two-layer methodology on a real-world safety-critical railway module.

The remainder of this paper is organised as follows: Section II provides the technical background and reviews related work. Section III describes the system architecture. Section IV details the verification methodology. Section V presents results and discusses limitations. Section VI concludes with recommendations for future work and methodology generalisation.

II. Background and Related Work

A. Legacy System and Prior Work

The target system is a microcomputer-controlled anti-skid (wheel-slide protection, WSP) system for urban rail vehicles [5]. Its test-diagnosis module uses an 8048 variant of the Intel MCS-48 family to perform non-real-time diagnostics: fault code storage and retrieval via 7-segment displays, and system self-testing with frequency sweep output.

Foundational reverse engineering by Ayoub [9] demonstrated the feasibility of FPGA emulation for the MCS-48 CPU. Leyer [10] extended this approach to the test-diagnosis board, targeting an Actel A3P1000 FPGA (flash-based, non-volatile) and integrating an STM32F401RET6 microcontroller for enhanced logging. However, the resulting Rev 0.0 prototype contained critical hardware defects—approximately 40 FPGA pins required re-soldering, the STM32 debug interface was blocked by a pin assignment error, and the Non-Volatile Random Access Memory (NVRAM) control logic contained a functional bug [11].

B. Verification Strategies for Retrofits

Demonstrating functional equivalence in safety-critical systems requires combining multiple verification approaches [6], [12]:

- **Black-box testing:** Comparing system outputs for identical inputs, including HIL testing [8], [13].
- **White-box analysis:** Simulation, static analysis, fault injection analysis [14], and structural testing when internal details (e.g., HDL code) are available.
- **Formal methods:** Mathematical equivalence checking [7], providing high assurance at the cost of substantial modelling effort and domain expertise.
- **Combined approaches:** Practice typically dictates layered strategies [15], [16], using white-box techniques for component verification and black-box methods for system validation.

C. Industry Precedents

Rail industry retrofits validate this approach. The Wabtec/SNCF adaptive wheel-slide protection project [17] achieved certification against EN 15595 [18] through HIL testing. Fleet-level electronics upgrades such as the UK Class 455 [19] and Greater Anglia Class 317 [20] further demonstrate the economic rationale for modernising safety-critical rolling stock [1].

D. Applicable Standards

The standards framework includes: IEC 61508 [4] (safety lifecycle, Safety Integrity Level (SIL)), EN 50129 [21] (hardware safety), EN 50716 [22] (software/HDL modification), EN 50155 [23] (environmental requirements), and IEEE 1012 [24] (V&V process). Using FPGAs in safety-critical applications introduces certification considerations including tool confidence, verification depth, and IP core validation [3], [25].

III. System Architecture

The retrofitted system employs a modular, four-Printed Circuit Board (PCB) architecture (Fig. 1) designed to enhance isolation, testability, and maintainability.

A. Base Board (FPGA Core)

The central hub houses the Actel A3P1000-PQG208 FPGA, chosen for its flash-based non-volatile configuration (inherent resistance to soft errors compared to SRAM alternatives [25]). The FPGA runs an MCS-48 emulation core (adapted from the OpenCores `t48_core`) executing the original 2 kB firmware binary from external flash ROM, thereby preserving the legacy control logic. In functional safety terms, this binary constitutes Software of Unknown Pedigree (SOUP), justified for integration via the IEC 61508 "Proven-in-Use" argument based on decades of safe field operation. Key peripherals emulated in VHDL include:

- Memory interfaces to Flash ROM (S29AL016J) and NVRAM (CY14B104NA) for persistent fault code storage.
- 7-segment display driving logic via Binary-Coded Decimal (BCD)-to-7-segment decoders (CD4511BE).
- A digital frequency sweep generator, replacing the legacy analog voltage-controlled oscillator (VCO). The generator employs a division counter logic that translates the 6 MHz system clock to an output frequency f_{out} according to the relation:

$$f_{\text{out}}(i) = \frac{f_{\text{clk}}}{2 \cdot N_{\text{div}}(i)}, \tag{1}$$

  where $f_{\text{clk}} = 6\,\text{MHz}$, and $N_{\text{div}}(i)$ is a clock division factor retrieved from a lookup table (LUT) implementing a non-linear sweep profile stepped by an internal 600 Hz clock. The division factor N_{div} varies between 1935 (yielding $f_{\text{out}} \approx 1550.4\,\text{Hz}$) and 3529 (yielding $f_{\text{out}} \approx 850.1\,\text{Hz}$), covering the required diagnostic range of 850 Hz to 1550 Hz across 33 steps (up-sweep) and 65 steps (down-sweep).
- An Serial Peripheral Interface (SPI) master for diagnostic data transmission to the STM32 subsystem.
- Watchdog interaction logic with an independent external hardware watchdog.

The board also integrates power regulation (TPS73615 and TPS73633), brown-out detection (Brown-Out Detection (BOD)) supervision via TPS3307, and a Joint Test Action Group (JTAG) interface for programming and boundary-scan testing.

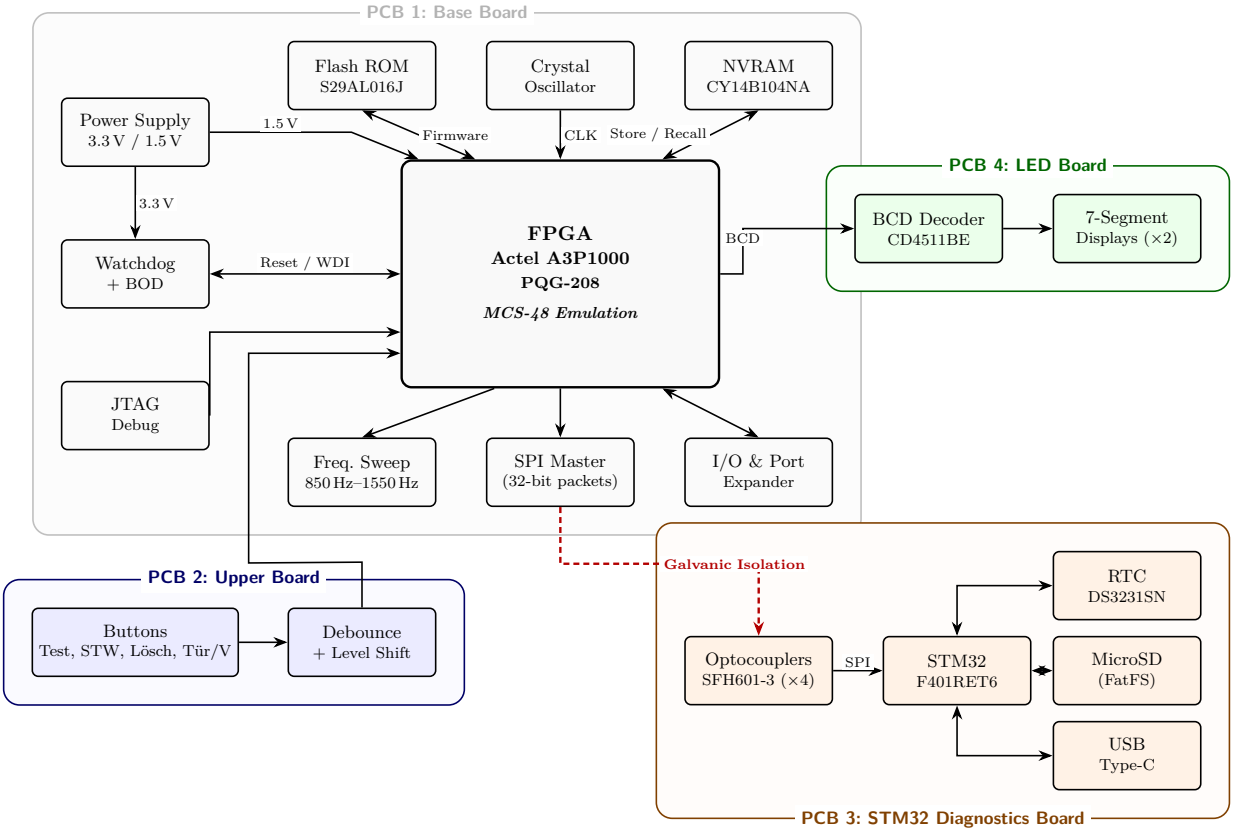


Fig. 1. High-level block diagram of the retrofitted test-diagnosis system. Four PCB modules are shown: the Base Board with the FPGA core (grey), the Upper Board with operator controls (blue), the LED Board (green), and the galvanically isolated STM32 Diagnostics Board (orange). The dashed red line marks the safety isolation boundary.

B. Upper Board (Legacy Interface)

This module interfaces with the legacy system's mechanical buttons (Test, Control [STW], Clear [Lösch], Door/Speed [Tür/V]) through debouncing and level-shifting circuitry, preserving the original user interaction paradigm.

C. STM32 Diagnostics Board

A dedicated STM32F401RET6 (Arm Cortex-M4) provides enhanced logging capabilities, *intentionally decoupled* from the safety-critical core through galvanic isolation. Four optocouplers (SFH601-3) isolate the SPI communication link, with separate isolated DC/DC converters maintaining the power barrier. To ensure reliable signal transmission across the isolation boundary, the FPGA's emulated SPI master clock frequency is set to $f_{\mathrm{SPI}} = 100\,\mathrm{kHz}$ by division of the 6 MHz system clock (30 clock cycles per half-period). Given that the SFH601-3 optocoupler has a typical propagation delay of $t_{\mathrm{prop}} \approx 3\,\mu\mathrm{s}$, this configuration leaves a timing margin of:

$$t_{\mathrm{margin}} = t_{\mathrm{half}} - t_{\mathrm{prop}} = 5\,\mu\mathrm{s} - 3\,\mu\mathrm{s} = 2\,\mu\mathrm{s}, \qquad (2)$$

which guarantees error-free signal transmission under typical operating temperatures. The subsystem features an Real-Time Clock (RTC) (DS3231SN) with battery backup for timestamped logging, a Secure Digital I/O (Secure Digital Input/Output (SDIO)) interface for MicroSD persistent storage, and Universal Serial Bus (USB) Type-C for data retrieval. While providing rich diagnostic data, the introduction of USB and SDIO interfaces expands the system's attack surface, necessitating future cybersecurity assessment aligned with IEC 62443 [26]. Similar diagnostic and logging architectures have been surveyed in rolling stock applications [27].

D. LED Display Board

Two Kingbright SC03-12EWA 7-segment displays, driven by CD4511BE decoders, provide visual fault code and self-test feedback to operators.

E. Safety-Relevant Design Features

Key safety features include: (1) an independent hardware watchdog forcing system reset upon FPGA hang detection, (2) BOD supervision on all critical power rails (5 V, 3.3 V, 1.5 V), (3) galvanic isolation preventing fault propagation from non-safety (STM32) to safety (FPGA) domains, and (4) retention of legacy built-in self-test capabilities.

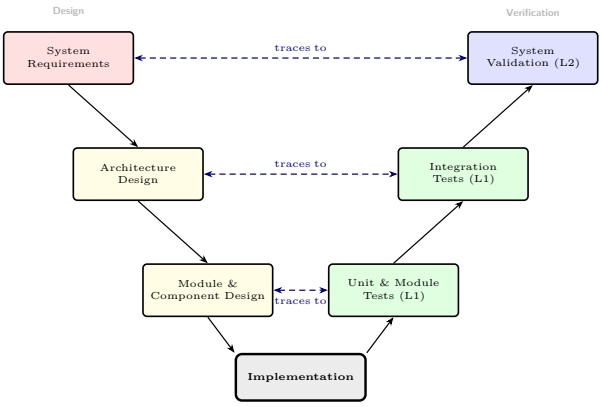


Fig. 2. Mapping of the two-layer verification strategy to the V-model. Layer 1 (green, L1) covers deterministic component and interface tests; Layer 2 (blue, L2) covers system-level black-box validation.

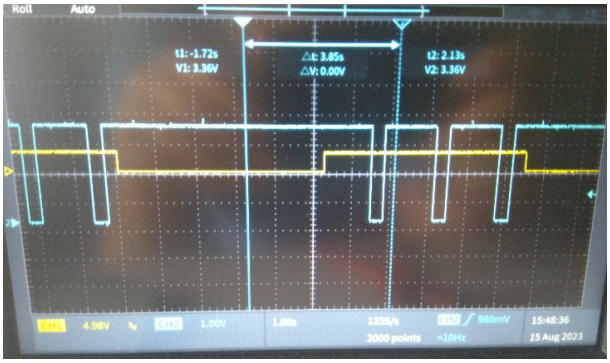


Fig. 3. Oscilloscope capture from Layer 1 verification showing the watchdog circuit timing. CH1 (yellow): FPGA servicing pulses. CH2 (blue): watchdog reset response.

IV. Verification Methodology

A. Two-Layer Verification Strategy

The verification strategy combines structural (white-box) component testing with functional (black-box) system validation, structured around the V-model [4], [22] and IEEE 1012 [24]. In software development, combining white-box and black-box testing throughout the V-model is standard practice; this work adapts and maps these complementary software engineering principles specifically to a hardware/firmware retrofit. This structured, two-layer design is necessary because verifying a new hardware core (the FPGA) while validating its functional equivalence against a legacy microcontroller's behaviour introduces both physical and logical integration challenges that cannot be resolved by a single testing paradigm. As shown in Fig. 2, the methodology maps to the right side of the V-model through two layers:

- **Layer 1 (Component & Interface Tests):** maps to *unit and integration testing*—deterministic, white-box verification proving that individual design artifacts (VHDL modules, hardware interfaces) are correctly implemented against their technical specifications.
- **Layer 2 (System-Level Tests):** maps to *system and acceptance testing*—black-box validation of the fully integrated system against original functional requirements (legacy system behaviour).

This combination—structural rigour at the component level (Layer 1) and functional fidelity at the integrated system level (Layer 2)—provides broader verification coverage than either technique in isolation.

B. Layer 1: Deterministic Component Verification

Layer 1 employed three complementary techniques:

1) VHDL Simulation: Extensive pre- and post-synthesis simulations using ModelSim ME verified the FPGA design prior to hardware deployment. Dedicated testbenches provided controlled stimuli to verify: the emulated MCS-48 core behaviour, memory interface timing against component datasheet specifications (Flash ROM, NVRAM), peripheral logic (display latching, button handling, frequency sweep, SPI master), and watchdog interaction (Fig. 3).

A critical finding during simulation was the discovery of the NVRAM control logic bug: the `NVRAM_ble` signal (active-low chip enable) was erroneously held at logic high in the inherited Rev 0.0 design, effectively disabling the non-volatile memory entirely [11]. This defect had remained undetected through previous development phases because the NVRAM was only exercised during fault storage and recall sequences—functions that were never systematically tested in isolation prior to this work. The fix was straightforward (permanently driving `NVRAM_ble` low), but its discovery illustrates the diagnostic value of dedicated component-level testbenches in catching design faults that persist through integration.

2) Boundary Scan (JTAG): IEEE 1149.1 boundary scan confirmed FPGA identification (A3P1000), successful bitstream programming, and I/O pin connectivity. This technique proved particularly valuable for diagnosing the approximately 40 FPGA pins that required re-soldering: boundary scan could rapidly identify open or bridged connections without requiring functional firmware execution.

3) Hardware Timing Measurements: Oscilloscope measurements on the assembled Rev 1.0 hardware validated critical real-world timing parameters: watchdog pulse periodicity and the frequency sweep profile (850 Hz to 1550 Hz). These measurements serve as the bridge between simulation-proven logical correctness and physical-world behaviour, confirming that the actual RC networks, crystal tolerances, and signal integrity meet design expectations.

C. Layer 2: Fixture-Based System Validation

Layer 2 validated the fully integrated system using a dedicated hardware test fixture ("Testkartentester") [28]. This layer addresses a fundamental limitation of Layer 1: simulation and hardware measurements verify individual components against their specifications, but cannot con-

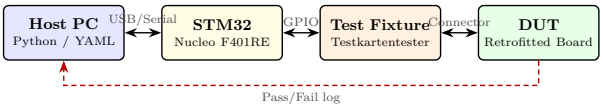


Fig. 4. Layer 2 test infrastructure. The PC orchestrates YAML-defined scenarios via an STM32 Nucleo board and the dedicated test fixture, which stimulates and monitors the Device Under Test (DUT).

firm that the complete system reproduces the legacy behaviour that operators and maintenance personnel rely on.

Test Infrastructure: The fixture (Fig. 4) provides physical connections via the main 64-pin system connector and interfaces with a host PC through an STM32 Nucleo-F401RE board. A Python application orchestrates test execution in a semi-automated workflow: it drives General-Purpose Input/Output (GPIO) pins to simulate fault conditions and system inputs, prompts the operator for manual checks (e.g., verifying 7-segment display codes visually), and logs pass/fail outcomes with timestamps. Each test scenario follows a four-phase cycle: (1) stimulus application via pin control, (2) settling delay (typically 3 s to match legacy button timing), (3) expected-output assertion, and (4) state cleanup via the Lösch (clear) button.

YAML-Based Scenarios: Test procedures were defined in YAML format, specifying: input pin stimuli (`pin_sets`), expected display codes (`expected_err_code`), expected stored fault sequences (`expected_err_seq`), and required tester actions. This format ensures repeatability and provides implicit traceability to legacy requirements [5].

Coverage: The test suite covered: (1) complete fault memory cycles (trigger → display → store → recall → clear) for all documented fault codes, (2) self-test activation and frequency sweep, (3) button interaction equivalence, (4) power-cycle stability, and (5) diagnostic SPI transmission to the STM32 logger.

Listing 1 illustrates a representative two-step test scenario for a "Geber 3" (sensor fault on axle card 3). The first step triggers the fault via `pin_sets`, asserts the expected immediate display code (`31`) and stored fault sequence (`03 31 09`). The second step verifies recall via the STW button and prompts the tester to clear the fault memory, confirming a complete storage-recall-clear cycle.

D. EN 50716 Modification Compliance

As a retrofit constitutes a modification, compliance with EN 50716 [22] was addressed through: *impact analysis* (modification confined to hardware; safety-critical firmware unchanged), *change verification* (the two-layer methodology itself), *regression testing* (Layer 2 scenarios as a comprehensive regression suite), and *configuration management* (Git version control for all artifacts).

Listing 1. Example YAML test scenario: Geber 3 fault memory cycle.

```yaml
# Geber 3 fault simulation (two-step sequence)
- name: Geber 3 - I
  description: >-
    Simulate sensor fault on axle 3. Wait 3 s for
    error display, then hold STW to verify sequence.
  pin_sets: {18z: 1, 24d: 0}
  expected_err_code: "31"
  expected_err_seq: "03␣31␣09"
  del: False

- name: Geber 3 - II
  description: Hold STW and verify stored sequence.
  pin_sets: {}
  expected_err_code: "95"
  expected_err_seq: "03␣31␣09"
  del: True   # Prompt tester to press 'Loesch'
```

V. Results and Discussion

A. Functional Equivalence Demonstration

Functional equivalence of the safety-critical core was demonstrated through three converging lines of evidence:

1) **Logical Correctness (VHDL Simulation):** After correcting the NVRAM control logic bug (`NVRAM_ble` was erroneously held high, disabling the chip [11]), all Layer 1 simulations confirmed correct memory interface timing, peripheral logic behaviour, and watchdog interaction.
2) **Physical Correctness (Hardware Measurement):** Post-rework oscilloscope measurements (Fig. 3) confirmed critical timing parameters, including a watchdog service period of $\Delta t = 3.85\,\mathrm{s}$ at $V = 3.36\,\mathrm{V}$ (expected $\approx 5\,\mathrm{s}$ from the VHDL test model; the shorter measured period is attributable to the voltage-dependent RC time constant of the external watchdog circuit) and a validated frequency sweep from 850 Hz to 1550 Hz. This phase included extensive rework—approximately 40 FPGA pins required re-soldering for stable operation [11].
3) **Behavioural Equivalence (System Scenario Testing):** All Layer 2 test scenarios passed after resolving integration bugs. The test suite covered all 20 documented fault conditions (five categories—sensor fault, short circuit, wire break, monitoring timeout, and watchdog—across four axle cards), including complete fault memory cycles (trigger → display → store → recall → clear), the full self-test sequence with frequency sweep, all four operator button interactions (Test, STW, Lösch, Tür/V), and power-cycle stability across multiple iterations.

The convergence of these three independent lines of evidence—each addressing a distinct aspect of correctness (logical, physical, behavioural)—provides robust assurance of functional equivalence for the safety-critical core. Because this methodology relies on generic V-model principles and black-box I/O behaviour rather than system-specific knowledge, it can be adapted to other IEC 61508-compliant hardware retrofits.

Table I provides a quantitative summary of all verification activities and their outcomes.

TABLE I
SUMMARY OF VERIFICATION RESULTS ACROSS BOTH LAYERS.

Test Category	Layer	Method	Result
Flash & NVRAM timing	L1	ModelSim	Pass
Periph. logic (disp/sweep)	L1	ModelSim	Pass
Watchdog interaction	L1	ModelSim	Pass
HW timing: watchdog	L1	Oscilloscope	3.85 s
HW timing: sweep	L1	Oscilloscope	850 Hz–1550 Hz
JTAG scan	L1	Libero SoC	Pass
Faults (20 codes)	L2	Fixture/YAML	20/20
Buttons (4 cases)	L2	Fixture/YAML	4/4
Self-test	L2	Fixture/YAML	Pass
Power stability	L2	Fixture/YAML	Pass
SPI (32-bit)	L2	Fixture/USB	Pass*
SD card	L2	Firmware	Pass†

*Originally truncated to 16 bits; resolved by resizing the SPI reception buffer configuration to 32 bits (4 bytes) in the STM32 firmware.
†Originally blocked due to filesystem mounting delays; resolved by mounting once at initialization and optimizing the main logging loop.

TABLE II
TAXONOMIC CLASSIFICATION OF DISCOVERED INTEGRATION BUGS.

ID	Class	Description	Layer	Instrument	Status
B1	Logical	NVRAM chip enable (`NVRAM_ble`) held high	L1	Simulation	Resolved
B2	Layout	STM32 SWD debug pins routed incorrectly	L1	Boundary-Scan	Reworked
B3	Assembly	Approx. 40 FPGA pins open or shorted	L1	JTAG / Scope	Reworked
B4	Firmware	FatFS timing starvation blocking SPI Rx	L2	Fixture/YAML	Resolved
B5	Interface	SPI payload truncation (16-bit vs 32-bit buffer)	L2	Fixture/USB	Resolved

Fig. 5 provides direct visual evidence of functional equivalence for the frequency sweep—a key anti-skid diagnostic function. The thin trend lines in both captures show close agreement between the original and emulated sweep profiles across the full 850 Hz–1550 Hz range.

B. Verification Coverage and the Diagnostic Blind Spot

The empirical validation of the retrofitted system on the target hardware revealed five distinct integration bugs across the hardware, firmware, and logical design domains, as categorized in Table II. An analysis of these defects highlights the complementary nature of the two-layer verification framework. Rather than introducing a novel testing paradigm, this structure applies the standard software engineering V-model principle—where structural component-level tests are followed by functional system-level validation—specifically to a hardware/firmware retrofit. The results empirically demonstrate that neither verification layer alone is sufficient to ensure safety-critical functional equivalence.

Structural verification and boundary-scan diagnostics (Layer 1) successfully isolated the NVRAM chip-enable error (B_1), the incorrect SWD pin routing (B_2), and the FPGA solder joint assembly defects (B_3). However, because Layer 1 lacks a dynamic runtime integration environment, it could not expose firmware-level timing starvation (B_4) or SPI buffer truncation issues (B_5).

Conversely, system-level validation using the dedicated test fixture (Layer 2) stimulated the system through its external connector pins and successfully captured the runtime integration errors B_4 and B_5. However, Layer 2 is constrained by observational limits. For example, during functional testing, the system's internal RAM cache temporarily satisfied transient read/write commands, rendering the disabled NVRAM (B_1) latent and invisible at the external board boundary. Similarly, the physical assembly faults (B_3) caused intermittent system crashes that were impossible to isolate or root-cause without Layer 1 boundary-scan tools.

Thus, the testing layers exhibit a mutually reinforcing relationship: Layer 1 verifies structural correctness and exposes latent bugs that escape functional testing, while Layer 2 validates real-time dynamic behaviour and exposes timing or protocol mismatches that cannot be captured in component simulations. Their convergence is essential for demonstrating absolute functional equivalence in safety-critical retrofits.

TABLE III
STANDARDS COMPLIANCE MAPPING FOR THE TWO-LAYER VERIFICATION METHODOLOGY.

Standard	Scope	Layer
IEC 61508 [4]	Safety lifecycle, SIL	L1 + L2
EN 50129 [21]	HW safety evidence	L1 + L2
EN 50716 [22]	SW/HDL modification	L1 + L2
EN 50155 [23]	Environmental req.	Future
EN 61373 [29]	Shock & vibration	Future
IEEE 1012 [24]	V&V process	L1 + L2

C. Safety Compliance

Table III summarises how the methodology maps to applicable standards. Layer 1 tests trace to HDL module requirements from reverse engineering; Layer 2 YAML scenarios trace to legacy functional requirements [5]. The generated artifacts under Git version control establish the evidence base for formal safety assessment. For certification, these implicit traceability links would need to be formalised into explicit requirements-to-test matrices.

D. Diagnostic Subsystem Validation

After re-routing the STM32 Serial Wire Debug (SWD) pins and debugging the firmware state machine [11], diagnostic data routing was validated via USB serial streaming, confirming isolation integrity, error-free 32-bit SPI packet reception, and robust SD logging.

E. Resolved Integration Issues

The discovery of the following integration issues during Layer 2 testing validates the necessity of the scenario-based approach; these faults would not have been detected

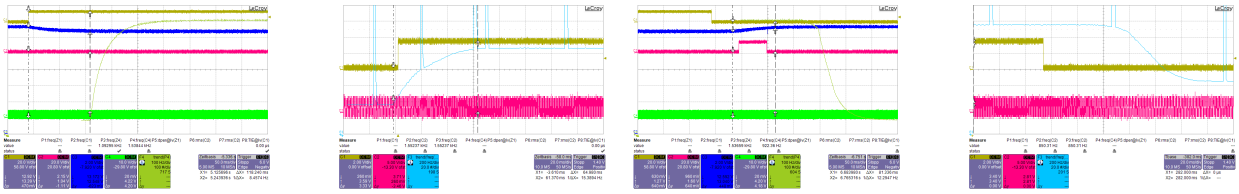


Fig. 5. Frequency sweep comparison: original system (left half of each panel, LeCroy oscilloscope) vs. FPGA emulation (right half). Left panel: ascending sweep (850 Hz–1550 Hz). Right panel: descending sweep (1550 Hz–850 Hz). The thin trend lines confirm matching frequency-response curves.

by Layer 1 simulation alone. To ensure absolute safety integrity, these issues were subsequently resolved:

- **SD Card Storage (FatFS Timing Starvation):** The FatFS implementation on the STM32 initially caused firmware timing starvation during block write attempts to the MicroSD card. Root-cause analysis revealed that the firmware unconditionally mounted and unmounted the filesystem volume on every write loop cycle, which introduced a 50 ms to 200 ms blocking overhead. This file system overhead, combined with blocking delays in the main execution thread, starved the SPI receive interrupt re-arming window, resulting in data overruns. This was resolved by mounting the SD card once at initialization, removing the loop-level mounting calls, and reducing the logging frequency.
- **SPI Payload Truncation and Alignment:** The VHDL SPI master correctly transmits a 32-bit diagnostic frame (consisting of a 2-bit prefix, 6-bit button state, and 24-bit error code). However, the STM32 firmware was initially misconfigured to receive only 16 bits per transaction (`HAL_SPI_Receive_IT` was called with a buffer size of 2). This mismatch truncated the payload and left the remaining 16 bits unread. The leftover clock pulses triggered a receiver buffer overrun (`OVR` flag in the SPI status register) and caused persistent frame misalignment on subsequent transfers. Correcting the STM32 RX buffer size configuration to 32 bits (4 bytes) resolved the truncation and misalignment.
- **Residual Scope Gaps:** Formal methods (e.g., model checking, equivalence checking), quantitative structural coverage metrics (statement/branch coverage for VHDL), accredited environmental testing (EN 50155 [23], EN 61373 [29]), and full HIL integration with the complete braking system were beyond project scope. These activities are required for formal safety certification and are identified as future work.

Crucially, both the SPI bug and the FatFS issue reside exclusively in the non-safety-critical diagnostic subsystem, which is galvanically isolated from the FPGA core. They do not affect the equivalence demonstration of the safety-critical functions verified through Layers 1 and 2 (see Sections IV-B and IV-C). This architectural separation—enforced by optocoupler isolation and independent power domains—ensures that diagnostic subsystem faults cannot propagate to the safety-critical domain. Board-level retrofits also align with UIC findings [1] that refurbishment reduces embodied CO_2 compared to new manufacture.

VI. Conclusion

This paper has proposed and validated a structured, two-layer verification methodology designed specifically for demonstrating functional equivalence in safety-critical FPGA retrofits by mapping standard V-model testing principles to the hardware/firmware integration domain. The central insight is that neither white-box simulation nor black-box system testing alone provides adequate coverage for modernising legacy safety hardware. Simulation cannot expose physical integration defects, while system-level testing lacks the granularity to isolate fundamental logic errors. By orchestrating a layered convergence of evidence—grounded in the V-model and IEEE 1012—the methodology establishes the rigorous diagnostic depth required to definitively prove functional equivalence with the original legacy system.

The practical realities of stabilising an inherited prototype—necessitating the re-soldering of roughly 40 FPGA pins alongside critical firmware interventions—highlight an indispensable lesson for hardware retrofits: legacy migration efforts must institutionalise a dedicated hardware stabilisation phase, and design-for-debuggability must be a primary, not secondary, requirement. These structural findings transcend the anti-skid domain, offering a transferrable blueprint for similar safety-critical legacy-to-FPGA modernisation strategies.

Future work: Immediate priorities involve correcting the STM32 firmware's SPI receiver buffer configuration to 32 bits and resolving the FatFS timing starvation with a non-blocking queue. Long-term objectives encompass full HIL integration [8], [13], rigorous cybersecurity assessments compliant with IEC 62443 [26], and the transition to professionally manufactured PCBs to endure accredited environmental certification (EN 50155 [23], EN 61373 [29]). Ultimately, this research paves the way for formalising the proposed verification architecture into a generalised, certifiable framework for the sustainable modernisation of IEC 61508-compliant railway electronics.

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